Decent Coaching Classes

VIII SB

Mind Your Math VIII

ANSWER KEY **Marks: 30**

Q.1 (A) **(4)**

(1) (C) $\sqrt{2}$ (2) (B) 115° (3) (C) 2 (4) (A) 2 : 1

Hint: (2) interior angles on the same side of the transversal are supplementary: 180° − 65° = 115°. (3) $125^{2/3} = (5^3)^{2/3} = 5^2 = 25$.

Q.1 (B) **(4)**

(1) $17^{1/3}$

(2) $-5/6 > -7/8$ [$-5/6 = -20/24$ and $-7/8 = -21/24$; $-20/24 > -21/24$]

(3) 8 angles

(4) True

Q.2 (any four, 2 marks each) **(8)**

(1) $(x + 5)(x - 3) = x^2 + (5 - 3)x + (5)(-3) =$ **$x^2 + 2x - 15$**

(2) $x^2 + 7x + 12 = x^2 + 3x + 4x + 12 = x(x + 3) + 4(x + 3) =$ **$(x + 3)(x + 4)$**

(3) $y \propto x$, so $y = kx$. $12 = k \times 3$, so **$k = 4$**. Equation: $y = 4x$. When $x = 7$, $y = 4 \times 7 =$ **28**.

(4) Diagonals of a rhombus bisect each other at right angles. Half-diagonals = 12 cm and 5 cm. $\text{Side}^2 = 12^2 + 5^2 = 144 + 25 = 169$, so **side = 13 cm**.

(5) Corresponding angles are congruent: $3x + 10 = 2x + 30$, so $3x - 2x = 30 - 10$, so **$x = 20$**.

(6) The centroid divides the median in the ratio 2 : 1, so l(AG) = 2 × l(GD). 8 = 2 × l(GD), so **l(GD) = 4 cm**. l(AD) = l(AG) + l(GD) = 8 + 4 = **12 cm**.

Q.3 (any two, 3 marks each) **(6)**

(1) $(2m - 3)^3 = (2m)^3 - 3(2m)^2(3) + 3(2m)(3)^2 - (3)^3$

$= 8m^3 - 3(4m^2)(3) + 3(2m)(9) - 27$

$=$ **$8m^3 - 36m^2 + 54m - 27$**

(2) Number of days is inversely proportional to number of workers, so (workers) × (days) = k. k = 12 × 15 = 180. For 20 workers: 20 × d = 180, so **d = 9 days**.

(3) Brokerage from one party = 6,00,000 × 2/100 = ₹ 12,000. Total brokerage = 12,000 + 12,000 = **₹ 24,000**. Amount received by seller = 6,00,000 − 12,000 = **₹ 5,88,000**.

Q.4 (any two, 4 marks each) **(8)**

(1) (a) $2x^2 - 5x - 3$: product of first and last coefficient = 2 × (−3) = −6; factors of −6 with sum −5 are −6 and +1.

$2x^2 - 6x + x - 3 = 2x(x - 3) + 1(x - 3) =$ **$(x - 3)(2x + 1)$** (2 marks)

(b) $27p^3 + 8q^3 = (3p)^3 + (2q)^3$ and $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$

$=$ **$(3p + 2q)(9p^2 - 6pq + 4q^2)$** (2 marks)

(2) Rough figure: Δ ABC has sides 4.5, 5, 6.5 cm and Δ ACD has sides 6.5, 5.5, 4 cm, sharing AC. Steps (1 mark each):

(i) Draw seg AC of length 6.5 cm.

(ii) With A as centre and radius 4.5 cm, and C as centre and radius 5 cm, draw arcs on one side of AC; they intersect at B. Join AB and BC.

(iii) On the opposite side of AC, with A as centre and radius 4 cm, and C as centre and radius 5.5 cm, draw arcs; they intersect at D.

(iv) Join AD and CD. □ ABCD is the required quadrilateral. (Check: AB = 4.5, BC = 5, CD = 5.5, DA = 4, AC = 6.5 cm.)

(3) Let the cost price be ₹ 100. Marked price = 100 + 20% of 100 = ₹ 120. Discount = 10% of 120 = ₹ 12. Selling price = 120 − 12 = ₹ 108. Profit = 108 − 100 = ₹ 8 on cost price ₹ 100, so **profit = 8%**.